WORKED SOLUTION

STEP II 2009, Question 7

Pure • Integration by Recognition

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q7

The key idea

Integrals like these look hopeless until you notice that they are *exactly* the derivative of something. The first result tells us what derivatives of $(x-a)^n e^{bx}\sqrt{1+x^2}$ look like: a power of $(x-a)$, the exponential, a cubic, all over $\sqrt{1+x^2}$. So each integral is really a matching exercise: read off n, a and b, and check the cubic agrees. When the polynomial in the numerator is a quartic rather than a cubic, we have to work out where the extra degree comes from.

Solution

The derivative

Differentiate the product of three factors, writing $\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{1+x^2} = \frac{x}{\sqrt{1+x^2}}$:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= n(x-a)^{n-1}e^{bx}\sqrt{1+x^2} + b(x-a)^n e^{bx}\sqrt{1+x^2} + (x-a)^n e^{bx}\frac{x}{\sqrt{1+x^2}} \\ &= \frac{(x-a)^{n-1}e^{bx}}{\sqrt{1+x^2}}\Big[\, n(1+x^2) + b(x-a)(1+x^2) + x(x-a)\,\Big],\end{aligned}$$

where we took out the common factor $(x-a)^{n-1}e^{bx}/\sqrt{1+x^2}$ (multiplying the first two terms through by $\sqrt{1+x^2}$ to put them over the common denominator). Expanding the bracket,

$$q(x) = bx^3 + (n+1-ab)x^2 + (b-a)x + (n-ab).$$

Since $b \neq 0$, the coefficient of x^3 is non-zero, so q really is a cubic. This is the required result. We will use it in the form

$$\int \frac{(x-a)^{n-1}e^{bx}\,q(x)}{\sqrt{1+x^2}}\,\mathrm{d}x = (x-a)^n e^{bx}\sqrt{1+x^2} + C. \qquad (*)$$

Part (i)

Matching the factors, we need $a = 4$, $b = 4$ and $n - 1 = 14$, so $n = 15$. With these values

$$q(x) = 4x^3 + (16-16)x^2 + (4-4)x + (15-16) = 4x^3 - 1,$$

which is exactly the cubic in the integrand. So by $(*)$

$$\int \frac{(x-4)^{14}e^{4x}(4x^3-1)}{\sqrt{1+x^2}}\,dx = (x-4)^{15}e^{4x}\sqrt{1+x^2} + C.$$

Part (ii)

Now the polynomial is a quartic, so it can't be q directly.

Ask yourself How can a quartic fit a formula that only produces cubics?

The power of $(x-a)$ in $(*)$ is ours to choose. If the quartic has a factor $(x-1)$, we can move it across to join $(x-1)^{21}$, leaving a cubic. Test $x = 1$: $12 - 1 - 11 = 0$, so $(x-1)$ is indeed a factor, and dividing out,

$$12x^4 - x^2 - 11 = (x-1)(12x^3 + 12x^2 + 11x + 11).$$

(You can also spot this as $(x^2-1)(12x^2+11)$.) The integrand is therefore

$$\frac{(x-1)^{22}e^{12x}(12x^3+12x^2+11x+11)}{\sqrt{1+x^2}}.$$

Take $a = 1$, $b = 12$, $n = 23$:

$$q(x) = 12x^3 + (24-12)x^2 + (12-1)x + (23-12) = 12x^3 + 12x^2 + 11x + 11,$$

which matches. Hence

$$\int \frac{(x-1)^{21}e^{12x}(12x^4-x^2-11)}{\sqrt{1+x^2}}\,dx = (x-1)^{23}e^{12x}\sqrt{1+x^2} + C.$$

Part (iii)

Again the polynomial is a quartic, but this time $x = 2$ gives $64 + 8 - 2 = 70 \neq 0$, so $(x-2)$ is not a factor and the trick from (ii) fails.

Ask yourself If $(x-2)$ isn't a factor, where does the extra degree come from?

Count degrees. The integrand's numerator has degree $6 + 4 = 10$, but $(*)$ with $(x-2)^6$ only gives degree $6 + 3 = 9$. An extra factor of x in the answer would raise the degree by one, so try differentiating xy, where y is the function with $a = 2$, $b = 4$, $n = 7$:

$$y = (x-2)^7e^{4x}\sqrt{1+x^2}, \qquad q(x) = 4x^3 + (8-8)x^2 + (4-2)x + (7-8) = 4x^3 + 2x - 1.$$

By the product rule $(xy)' = y + xy'$. Putting y over the denominator $\sqrt{1+x^2}$ as well,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(xy) &= \frac{(x-2)^7\mathrm{e}^{4x}(1+x^2)}{\sqrt{1+x^2}} + \frac{x(x-2)^6\mathrm{e}^{4x}(4x^3+2x-1)}{\sqrt{1+x^2}} \\ &= \frac{(x-2)^6\mathrm{e}^{4x}}{\sqrt{1+x^2}}\Big[(x-2)(1+x^2) + 4x^4 + 2x^2 - x\Big].\end{aligned}$$

Since $(x-2)(1+x^2) = x^3 - 2x^2 + x - 2$, the bracket is

$$x^3 - 2x^2 + x - 2 + 4x^4 + 2x^2 - x = 4x^4 + x^3 - 2,$$

exactly the quartic we want. Therefore

$$\int \frac{(x-2)^6\mathrm{e}^{4x}(4x^4+x^3-2)}{\sqrt{1+x^2}}\,\mathrm{d}x = x(x-2)^7\mathrm{e}^{4x}\sqrt{1+x^2} + C.$$

Checks

- Differentiate each answer and compare: this is the definitive check, and in each case it is just $(*)$ again (plus the product rule in (iii)).
- Quick degree count: differentiating an answer of polynomial degree d (ignoring the square root and exponential) gives a numerator of degree $d+2$. In (i), $15+2 = 14+3$; in (ii), $23+2 = 21+4$; in (iii), $8+2 = 6+4$.
- Numerically, Simpson's rule for the integral in (iii) over $[-1, 1.2]$ gives $-104.41424\ldots$, agreeing with $F(1.2) - F(-1)$ for $F(x) = x(x-2)^7\mathrm{e}^{4x}\sqrt{1+x^2}$ to about 13 significant figures; (i) and (ii) agree just as well.

Common slips

- Forgetting that n is one more than the power in the integrand: in (i) the answer has $(x-4)^{15}$, not $(x-4)^{14}$.
- In (ii), trying to match the quartic directly and concluding there is no solution, instead of first absorbing the factor $(x-1)$.
- Leaving out the constant of integration.

Beyond STEP

The method here is "integration by recognition": guess the shape of an antiderivative, differentiate it, and match coefficients. Turned into an algorithm, it becomes the Risch algorithm, which computer algebra systems use to decide whether an elementary function has an elementary antiderivative, and to find it if so. A related idea, guessing a solution of a given form with unknown coefficients, is the method of undetermined coefficients for linear differential equations, covered in Boyce and DiPrima, *Elementary*

Differential Equations. For more integrals in the STEP style, see Siklos, *Advanced Problems in Mathematics*.

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